

Experiment 3 — Mobile Cloud Services for Backend Support using Firebase

Step 1: Create Firebase Project

1. Open your browser and go to <https://console.firebase.google.com>
2. Log in with your **Google account**
3. Click **Create a Project**
4. Enter a project name (e.g., **MobileCloudApp**)
5. Click **Continue**
6. Disable Google Analytics (optional for beginners) or keep it enabled
7. Click **Create Project** and wait for setup to complete
8. Click **Continue** to open the project dashboard

 **Screenshot needed:** Firebase project dashboard showing your project name and "Hello, [Your Name]"

Step 2: Enable Authentication Service

1. In the left sidebar, click **Build** → **Authentication**
2. Click **Get Started**
3. Open the **Sign-in Method** tab
4. Click on **Email/Password** provider
5. Toggle **Enable** to ON for Email/Password
6. Keep **Email link (passwordless sign-in)** disabled
7. Click **Save**

 **Screenshot needed:** Sign-in providers list showing Email/Password with status "Enabled"

Step 3: Create Cloud Database (Firestore)

1. In the left sidebar, click **Build** → **Firestore Database**
2. Click **Create Database**
3. Select **Start in Test Mode** (for learning purposes)
4. Choose location: **asia-south1** (closest to India)
5. Click **Enable** and wait for the database to provision
6. Once ready, click **Start Collection**
7. Enter Collection ID: **users**
8. Click **Next**
9. Set Document ID to **Auto-ID** or type **user1**
10. Add the following fields:

- Field: **name** → Value: your name
- Field: **email** → Value: your email
- Field: **role** → Value: student

11. Click **Save**

 **Screenshot needed:** Firestore database view showing the users collection with your document and fields visible

Step 4: Enable Cloud Storage

1. In the left sidebar, click **Build** → **Storage**
2. Click **Get Started**

Note: Firebase Storage requires a Blaze (pay-as-you-go) plan upgrade to enable. If the upgrade prompt appears, you can skip the actual upload step and just screenshot the Storage landing page as evidence of reaching this step.

3. If accessible, click **Next**, choose default location, click **Done**
4. Click **Upload File**
5. Select any image (.jpg/.png) or PDF document from your system
6. Wait for upload to complete

 **Screenshot needed:** Storage section showing either the upgrade prompt or the uploaded file in the storage bucket

Step 5: Mobile Cloud Architecture Analysis

Write the following analysis in your journal:

Role of Authentication Service: Firebase Authentication provides secure user login without building a custom auth system. Email/Password sign-in is managed entirely by Firebase, reducing development effort and security risk.

Real-Time Data Synchronization: Firestore automatically syncs data across all connected clients in real time. Any change in the database is instantly reflected on all devices using the app without manual refresh.

Storage Usage for Mobile Apps: Firebase Storage handles binary files like images, audio, and documents, decoupling file storage from the app logic. This is essential for mobile apps that frequently upload and retrieve media content.

Backend Scalability: Firebase automatically scales backend infrastructure based on usage. Developers don't need to provision or manage servers, making it ideal for growing mobile applications.

The architecture for this setup follows a standard Mobile Cloud pattern: Mobile App → Firebase Auth → Firestore DB / Firebase Storage → Google Cloud backend. No further elaboration needed here.

Final Summary:

- Open Firebase and create project
- Search Authentication and add Sign-in method Email/Password and Google
- Search Firestore and create database in test mode
- Click start collection and add fields and values
- Search Cloud Storage (unavailable for free plan)